

空间的层级与信息的边界

Hierarchy of Spaces and Boundary of Information

从纯笛卡尔空间到六维复空间的维度进化

Dimensional Evolution from Pure Cartesian Space to 6-Dimensional Complex Space

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摘要 / Abstract

中文

人类对物理世界的描述始于笛卡尔空间 (x, y, z) 。本文在角动量网络几何学 (ANG-TOE) 框架中，建立了一个从最不完整到最完整的空间层级体系。我们分析四个层级的空间：

1. 纯笛卡尔空间 (3D)：人类最熟悉的静态位置空间。
2. 感知时频相空间 (6D)：人类可读的完整读数空间。
3. 物理角动量空间 (6D，即六个实矢量分量)：宇宙物理实在的闭包。
4. 完整图空间 (12D 实空间 = 六维复空间 $\mathbb{C}^6$)：物理本体与感知投影的无损并置。

我们严格证明：物理角动量空间由六个实矢量分量 $(L_x, L_y, L_z, S_x, S_y, S_z)$ 张成。六维复空间 $\mathbb{C}^6$ 是完整图空间的自然表示——三个轨道分量和三个自旋分量各用一个复数表

示，实部编码物理角动量，虚部编码对应的感知投影读数。这是感知时频相空间的自然扩展，使得人类可以通过复数形式理解物理-感知对偶。

关键词：空间层级、维度缺失、笛卡尔空间、时频相空间、角动量空间、图空间、六维复空间

English

The human description of the physical world originates from Cartesian space (x, y, z) . Within the Angular-Momentum Network Geometry (ANG-TOE) framework, this paper constructs a hierarchical system of spaces ordered from the least complete to the most complete. Four levels are analyzed:

1. Pure Cartesian space (3D): the familiar static position space for human intuition.
2. Perceptual time-frequency-phase space (6D): the full human-readable measurement space.
3. Physical angular-momentum space (6D, spanned by six real vector components): the ontological closure of cosmic physical reality.
4. Full graph space (12-dimensional real space = 6-dimensional complex space $\mathbb{C}^6$): lossless juxtaposition of physical ontology and perceptual projection.

We rigorously demonstrate that the physical angular-momentum space is a manifold spanned by six real vector components $(L_x, L_y, L_z, S_x, S_y, S_z)$. The 6-dimensional complex space $\mathbb{C}^6$ serves as the natural representation of full graph space: each of the three orbital components and three spin components forms one complex number, where the real part encodes physical angular momentum and the imaginary part encodes the corresponding perceptual readings. As a natural extension of the time-frequency-phase space, this complex formulation enables an intuitive understanding of ontology-perception duality.

Keywords: Hierarchy of Spaces, Dimensional Deficiency, Cartesian Space, Time-Frequency-Phase Space, Angular-Momentum Space, Graph Space, 6-Dimensional Complex Space

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1. 引言：为什么笛卡尔空间不够用？

中文

人类对物理世界的空间描述始于笛卡尔坐标系 (x, y, z) 。三个互相垂直的轴，足够描述一个点在空间中的位置。然而，随着物理学逐渐深入，人们发现这个空间越来越不够用：

- 为了描述运动，我们引入了时间 t 。
- 为了描述波动，我们引入了频率 ω 。
- 为了描述量子关联，我们引入了相位 Φ 。
- 为了描述旋转，我们引入了角动量 $(\mathbf{L}, \mathbf{S})$ 。

笛卡尔空间本身没有改变，但物理学不断在它外面“添加”新的维度。这说明：纯笛卡尔空间从一开始就不完整。

本文系统性地分析从最不完整到最完整的空间层级，指出每一个层级的信息承载能力与信息缺失。特别地，我们给出两个核心定义：

- 物理角动量空间由六个实矢量分量构成。
- 六维复空间 $\mathbb{C}^6$ 是完整图空间的表示，其中实部编码物理角动量，虚部编码感知投影。

English

Human spatial description of nature began with the Cartesian coordinate system (x, y, z) . Three orthogonal axes are sufficient to mark the position of a point. As physics advanced, this basic framework proved increasingly inadequate:

- Time t was introduced to describe motion.
- Frequency ω was introduced to describe wave phenomena.
- Phase Φ was introduced to describe quantum coherence.
- Angular momentum $(\mathbf{L}, \mathbf{S})$ was introduced to describe rotation and spin.

Cartesian space itself remains unchanged, while new dimensions are appended externally. This indicates that pure Cartesian space is intrinsically incomplete from the outset.

This paper systematically analyzes four hierarchical space structures and clarifies the information-bearing capacity and information-loss pattern at each level. Two core definitions are emphasized:

1. The physical angular-momentum space consists of six real vector components.
2. The 6-dimensional complex space $\mathbb{C}^6$ represents full graph space, where real parts encode physical angular momentum and imaginary parts encode perceptual projections.

2. 纯笛卡尔空间：3D 截面的困境

2.1 定义 / Definition

中文

$$\mathcal{S}_3^{\text{Cart}} = \{x, y, z\}$$

English

$$\mathcal{S}_3^{\text{Cart}} = \{x, y, z\}$$

2.2 维度 / Dimension

中文：3

English: 3-dimensional

2.3 信息承载能力 / Information-Carrying Capacity

中文

纯笛卡尔空间仅能描述一个静态点的位置。它无法承载任何与“变化”、“关系”、“相互作用”相关的信息。

English

Pure Cartesian space can only describe the static position of a point. It cannot carry any information related to variation, correlation or physical interaction.

2.4 缺失的信息 / Missing Information

中文

缺失的维度	物理含义	缺失的后果
时间 t	演化方向	只能描述静态快照，无法描述运动、力、能量交换
频率 ω	振动的能量尺度	无法描述波、量子能级、温度、光谱
相位 Φ	链接网络的相干度	无法描述量子关联、拓扑荷、意识相变、纠缠
角动量 ($\mathbf{L}, \mathbf{S}$)	旋转与自旋	无法描述轨道运动、自旋、磁矩、量子数

English

Missing Dimension	Physical Meaning	Consequence of Deficiency
Time t	Direction of evolution	Only static snapshots are available; motion, force and energy exchange cannot be represented
Frequency ω	Energy scale of oscillation	Waves, quantum energy levels, temperature and spectra cannot be described
Phase Φ	Coherence of link network	Quantum correlation, topological charge, consciousness phase transition and entanglement are inaccessible
Angular momentum ($\mathbf{L}, \mathbf{S}$)	Rotation and spin	Orbital motion, spin, magnetic moment and quantum numbers cannot be captured

2.5 数学地位 / Mathematical Status

中文

$$\mathcal{S}_3^{\text{Cart}} \subset \mathcal{M}_6^{\text{percep}} \text{ 且 } \mathcal{S}_3^{\text{Cart}} \not\supset \mathcal{M}_6^{\text{phys}}$$

纯笛卡尔空间既不是物理本体空间的子集，也不是感知投影空间的闭包。它只是人类为了方便在草稿纸上画图而截取的低维截面——一个退化的坐标框架。

English

$$\mathcal{S}_3^{\text{Cart}} \subset \mathcal{M}_6^{\text{percep}} \text{ and } \mathcal{S}_3^{\text{Cart}} \not\subset \mathcal{M}_6^{\text{phys}}$$

Pure Cartesian space is neither a subset of physical ontological space nor the closure of perceptual projection space. It is merely a low-dimensional cross-section intercepted for human convenience, a degenerate coordinate frame.

3. 感知时频相空间：6D 可读界面

3.1 定义 / Definition

中文

$$\mathcal{M}_6^{\text{percep}} = \{x, y, z, t, \omega, \Phi\}$$

English

$$\mathcal{M}_6^{\text{percep}} = \{x, y, z, t, \omega, \Phi\}$$

3.2 维度 / Dimension

中文：6

English: 6-dimensional

3.3 信息承载能力 / Information-Carrying Capacity

中文

这是人类用仪器可读出的完整读数空间。它包含了位置、时间、频率和相位——足以描述人类能测量的全部物理量。

English

This is the complete measurement space accessible to human instruments. It contains position, time, frequency and phase, sufficient to represent all physically observable quantities.

3.4 坐标的物理依赖性 / Physical Dependence of Coordinates

中文

坐标	定义方式	物理依赖性
t （秒）	铯-133 原子跃迁频率	依赖于局域 Φ （原子钟修正）

x, y, z (米)	光在 1/299792458 秒内传播距离	依赖于局域光速 $c(\Phi)$
ω (赫兹)	事件重复速率的测量值	依赖于局域重联速率
Φ (相位)	信号相位的读数	依赖于链接网络相干度

English

Coordinate	Definition	Physical Dependence
t (second)	Transition frequency of Cs-133	Depends on local Φ (atomic clock correction)
x, y, z (meter)	Distance light travels in 1/299792458 seconds	Depends on local light speed $c(\Phi)$
ω (hertz)	Measured repetition rate of events	Depends on local reconnection rate
Φ (phase)	Observed signal phase	Depends on the coherence of the link network

3.5 缺失的信息 / Missing Information

中文

感知空间是投影空间，不是源空间。它丢失了物理本体的非对角耦合分量。

$\dim(\ker \mathcal{J}_\pi) = 6 - \text{rank}(\mathcal{J}_\pi) > 0$

丢失的信息	物理含义	丢失的后果
非对角角动量耦合	L 与 S 之间的干涉	无法区分“轨道运动”与“自旋”的深层耦合
全局拓扑相	链接网络的整体缠绕数	无法感知全局拓扑不变量
角动量对消残余	Axiom 0 的局域偏差	无法检测引力波、暗物质尾迹

English

Perceptual space is a projected image rather than the original source space.
Off-diagonal coupling components from physical ontology are lost:
 $\dim(\ker \mathcal{J}_\pi) = 6 - \text{rank}(\mathcal{J}_\pi) > 0$

Lost Information	Physical Meaning	Consequence of Loss
Off-diagonal angular-momentum coupling	Interference between L and S	Deep coupling between orbital motion and spin cannot be distinguished
Global topological phase	Global winding number of link network	Global topological invariants become unobservable
Residual angular-momentum offset	Local deviation from Axiom 0	Gravitational-wave signatures and dark-matter trails cannot be detected

3.6 数学地位 / Mathematical Status

中文

$$\mathcal{M}_6^{\text{percep}} = \mathcal{P}(\mathcal{M}_6^{\text{phys}})$$

即时频相空间是物理角动量空间在 Axiom IV 投影下的像。它是一个读数界面，不是宇宙的底层语法。

English

$$\mathcal{M}_6^{\text{percep}} = \mathcal{P}(\mathcal{M}_6^{\text{phys}})$$

Namely: the time-frequency-phase space is the image of physical angular-momentum space under the projection operator defined in Axiom IV. It serves as a measurement interface rather than the fundamental grammar of the universe.

4. 物理角动量空间：6D 实在闭包

4.1 定义 / Definition

中文

$$\mathcal{M}_6^{\text{phys}} = \{\mathbf{L}, \mathbf{S}\}$$

其中：

$\mathbf{L} = (L_x, L_y, L_z)$: 轨道角动量（三维实矢量）

$\mathbf{S} = (S_x, S_y, S_z)$: 自旋角动量（三维实矢量）

English

$$\mathcal{M}_6^{\text{phys}} = \{\mathbf{L}, \mathbf{S}\}$$

Where:

- $\mathbf{L} = (L_x, L_y, L_z)$: orbital angular momentum (3-dimensional real vector)

- $\mathbf{S} = (S_x, S_y, S_z)$: spin angular momentum (3-dimensional real vector)

4.2 维度 / Dimension

中文：6（实）

$$\mathcal{M}_6^{\text{phys}} = \mathbb{R}^6$$

坐标基底：

$$\mathbf{q} = (L_x, L_y, L_z, S_x, S_y, S_z) \in \mathbb{R}^6$$

编号	符号	物理含义	类型
1	L_x	轨道角动量 X 分量	实标量
2	L_y	轨道角动量 Y 分量	实标量
3	L_z	轨道角动量 Z 分量	实标量
4	S_x	自旋角动量 X 分量	实标量
5	S_y	自旋角动量 Y 分量	实标量
6	S_z	自旋角动量 Z 分量	实标量

English: 6 (real)

$$\mathcal{M}_6^{\text{phys}} = \mathbb{R}^6$$

Coordinate basis:

$$\mathbf{q} = (L_x, L_y, L_z, S_x, S_y, S_z) \in \mathbb{R}^6$$

Index	Symbol	Physical Meaning	Type
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1	L_x	X-component of orbital angular momentum	Real scalar
2	L_y	Y-component of orbital angular momentum	Real scalar
3	L_z	Z-component of orbital angular momentum	Real scalar
4	S_x	X-component of spin angular momentum	Real scalar
5	S_y	Y-component of spin angular momentum	Real scalar
6	S_z	Z-component of spin angular momentum	Real scalar

4.3 信息承载能力 / Information-Carrying Capacity

中文

这是 ANG-TOE 中宇宙物理实在的闭包。所有链接网络在此空间中演化（Axiom VI），所有角动量在此空间中归零（Axiom 0）。

English

This manifold is the ontological closure of cosmic reality within ANG-TOE. All link networks evolve inside this space (Axiom VI), and global angular momentum vanishes under Axiom 0.

4.4 完整物理量的表达 / Expressions of Basic Physical Quantities

中文

物理量	在 $\mathcal{M}_6^{\text{phys}}$ 中的表达
-----	--------------------------------------

质量 m	$m = \frac{\hbar}{R_5 c} \cdot \frac{\text{Link}}{\chi} \cdot \mathcal{F}$
能量 E	$E = \partial \mathbf{J} / \partial t$
力 $\mathbf{F}$	$\mathbf{F} = -\nabla \mathbf{J}$
时间 Δt	$\Delta t = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$

English

Physical Quantity	Expression within $\mathcal{M}_6^{\text{phys}}$
Mass m	$m = \frac{\hbar}{R_5 c} \cdot \frac{\text{Link}}{\chi} \cdot \mathcal{F}$
Energy E	$E = \partial \mathbf{J} / \partial t$
Force $\mathbf{F}$	$\mathbf{F} = -\nabla \mathbf{J}$
Time interval Δt	$\Delta t = \frac{\bar{\ell}}{\bar{\kappa} \cdot \omega_{\text{topo}}}$

4.5 信息完整性 / Information Completeness

中文

rank($\mathcal{M}_6^{\text{phys}}$) = 6

满秩。物理本体空间本身不含信息缺失。信息缺失只发生在投影到感知空间的过程中。

English

rank($\mathcal{M}_6^{\text{phys}}$) = 6

Full rank. No intrinsic information loss exists within the ontological space itself. Information loss only arises after projection into perceptual coordinates.

5. 完整图空间：12D 无损信息体与六维复空间

5.1 定义 / Definition

中文

为了在保持可读性的同时不丢失任何物理信息，必须同时保留物理本体和感知投影：

$$\mathcal{M}_{12}^{\text{graph}} = \mathcal{M}_6^{\text{phys}} \oplus \mathcal{M}_6^{\text{percep}}$$

等价地：

$$\mathcal{M}_{12}^{\text{graph}} = \{(\mathbf{q}, \mathcal{P}(\mathbf{q})) \mid \mathbf{q} \in \mathcal{M}_6^{\text{phys}}\}$$

English

To preserve full physical information while retaining measurement readability, both the ontological state and its perceptual projection must be stored together:

$$\mathcal{M}_{12}^{\text{graph}} = \mathcal{M}_6^{\text{phys}} \oplus \mathcal{M}_6^{\text{percep}}$$

Equivalently:

$$\mathcal{M}_{12}^{\text{graph}} = \{(\mathbf{q}, \mathcal{P}(\mathbf{q})) \mid \mathbf{q} \in \mathcal{M}_6^{\text{phys}}\}$$

5.2 维度 / Dimension

中文：12（实）= 六维复空间 $\mathbb{C}^6$

$$\mathcal{M}_{12}^{\text{graph}} \cong \mathbb{C}^6$$

English: 12 (real) = 6-dimensional complex space $\mathbb{C}^6$

$$\mathcal{M}_{12}^{\text{graph}} \cong \mathbb{C}^6$$

5.3 六维复空间的自然配对（物理-感知对偶） / Natural Pairing in $\mathbb{C}^6$ (Ontology-Perception Duality)

中文

复数编号	实部（物理本体）	虚部（感知投影）	物理含义
Z_1	L_x （轨道角动量 X）	x （空间 X 位置）	轨道-空间 X 耦合
Z_2	L_y （轨道角动量 Y）	y （空间 Y 位置）	轨道-空间 Y 耦合
Z_3	L_z （轨道角动量 Z）	z （空间 Z 位置）	轨道-空间 Z 耦合
Z_4	S_x （自旋角动量 X）	t （时间）	自旋-时间耦合

Z_5	S_y (自旋角动量 Y)	ω (频率)	自旋-频率耦合
Z_6	S_z (自旋角动量 Z)	Φ (相位)	自旋-相位相干耦合

六个复数:

- 三个轨道复数: Z_1, Z_2, Z_3 编码轨道自由度及其空间投影
- 三个自旋复数: Z_4, Z_5, Z_6 编码自旋自由度及其时频相投影

English

Complex Index	Real Part (Ontology)	Imaginary Part (Perception)	Physical Interpretation
Z_1	L_x (orbital-X)	x (spatial-X)	Orbit-space X coupling
Z_2	L_y (orbital-Y)	y (spatial-Y)	Orbit-space Y coupling
Z_3	L_z (orbital-Z)	z (spatial-Z)	Orbit-space Z coupling
Z_4	S_x (spin-X)	t (time)	Spin-time coupling
Z_5	S_y (spin-Y)	ω (frequency)	Spin-frequency coupling
Z_6	S_z (spin-Z)	Φ (phase)	Spin-phase coherence coupling

Six complex numbers:

- Three orbital complexes Z_1, Z_2, Z_3 encode orbital degrees-of-freedom together with their spatial projections
- Three spin complexes Z_4, Z_5, Z_6 encode spin degrees-of-freedom together with their time-frequency-phase projections

5.4 为什么六维复空间是自然的表示? / Why $\mathbb{C}^6$ Is the Natural Representation

中文

理由	解释
物理匹配	物理本体（实部）与感知投影（虚部）刚好成对出现，构成复平面上的“物理-感知对偶”
信息无损	从 $\mathbb{C}^6$ 中读取实部即得到物理本体，读取虚部即得到感知投影，两个方向都不丢失信息
投影算子	投影 $\mathcal{P}$ 在这个复空间中退化为“取虚部”操作： $\mathcal{P}(Z) = \text{Im}(Z)$
逆投影	逆投影 $\mathcal{E}$ 退化为“补实部”操作： $\mathcal{E}(X) = \text{Re}(Z)$

English

Reason	Explanation
Physical Matching	Ontology (real part) and perceptual projection (imaginary part) appear in pairs and form ontology-perception duality on the complex plane
Lossless Information	Reading real parts recovers physical ontology; reading imaginary parts recovers perceptual measurements, with no information loss in either direction
Projection Operator	Within this complex manifold, projection simplifies to taking the imaginary part: $\mathcal{P}(Z) = \text{Im}(Z)$
Inverse Projection	Inverse projection recovers the real-part ontology: $\mathcal{E}(X) = \text{Re}(Z)$

5.5 信息完整性 / Information Completeness

中文

$$\mathcal{P} \circ \mathcal{E}_{\text{graph}} = \mathbf{I}_6 \text{ 且 } \mathcal{E}_{\text{graph}} \circ \mathcal{P} = \mathbf{I}_6$$

映射是双射的。从图空间可以无损恢复物理本体，也可以无损恢复感知读数。

English

$$\mathcal{P} \circ \mathcal{E}_{\text{graph}} = \mathbf{I}_6 \text{ and } \mathcal{E}_{\text{graph}} \circ \mathcal{P} = \mathbf{I}_6$$

The mapping is bijective. Both the original ontological state and the perceptual measurement can be reconstructed without information loss from graph space.

5.6 信息恢复路径 / Information Reconstruction Path

中文

操作	从 $\mathcal{M}_{12}^{\text{graph}}$ 读取	结果
读实部	$(\mathbf{L}, \mathbf{S})$	物理本体（完整）
读虚部	$(x, y, z, t, \omega, \Phi)$	感知读数（完整）
读映射关系	$(\mathbf{q}, \mathcal{P}(\mathbf{q}))$	投影结构（完整）

English

Operation	Read from $\mathcal{M}_{12}^{\text{graph}}$	Result
Read real parts	$(\mathbf{L}, \mathbf{S})$	Complete physical ontology
Read imaginary parts	$(x, y, z, t, \omega, \Phi)$	Complete perceptual measurements
Read mapping relation	$(\mathbf{q}, \mathcal{P}(\mathbf{q}))$	Complete projection structure

6. 空间层级对比与汇总 / Comparison and Summary of Spatial Hierarchies

6.1 完整对比表 / Full Comparison Table

中文

层级	空间名称	实维度	复数表示	坐标	信息完整性
L1	纯笛卡尔空间	3	✗ 否	(x, y, z)	✗ 严重缺失

L2	感知时频相空间	6	✗ 否	$(x, y, z, t, \omega, \Phi)$	✗ 投影有损
L3	物理角动量空间	6	-	$(\mathbf{L}, \mathbf{S})$	☑ 物理闭包
L4	完整图空间	12	☑ 是 ($\mathbb{C}^6$)	$(\mathbf{q}, \mathcal{P}(\mathbf{q}))$	☑ 完整无损

English

Level	Space Name	Real-Dim	Complex Representation	Coordinates	Information Completeness
L1	Pure Cartesian Space	3	✗ No	(x, y, z)	✗ Severe deficiency
L2	Perceptual Time-Frequency-Phase Space	6	✗ No	$(x, y, z, t, \omega, \Phi)$	✗ Lossy projection
L3	Physical Angular-Momentum Space	6	-	$(\mathbf{L}, \mathbf{S})$	☑ Physically closed
L4	Full Graph Space	12	☑ Yes ($\mathbb{C}^6$)	$(\mathbf{q}, \mathcal{P}(\mathbf{q}))$	☑ Lossless complete

6.2 各层级信息缺失的具体内容 / Detailed Information-Loss Table

中文

缺失内容	L1	L2	L3	L4
时间演化	✗ 缺失	☑ 包含	☑ 包含	☑ 包含
频率/能量尺度	✗ 缺失	☑ 包含	☑ 包含	☑ 包含
相位相干	✗ 缺失	☑ 包含	☑ 包含	☑ 包含

轨道角动量	✗ 缺失	✗ 缺失	☑ 包含	☑ 包含
自旋角动量	✗ 缺失	✗ 缺失	☑ 包含	☑ 包含
角动量耦合	✗ 缺失	✗ 缺失	☑ 包含	☑ 包含
投影映射关系	✗ 缺失	✗ 缺失	✗ 缺失	☑ 包含
物理-感知对偶	✗ 缺失	✗ 缺失	✗ 缺失	☑ 包含

English

Missing Item	L1	L2	L3	L4
Time evolution	✗ Missing	☑ Included	☑ Included	☑ Included
Frequency / energy scale	✗ Missing	☑ Included	☑ Included	☑ Included
Phase coherence	✗ Missing	☑ Included	☑ Included	☑ Included
Orbital angular momentum	✗ Missing	✗ Missing	☑ Included	☑ Included
Spin angular momentum	✗ Missing	✗ Missing	☑ Included	☑ Included
Angular-momentum coupling	✗ Missing	✗ Missing	☑ Included	☑ Included
Projection mapping relation	✗ Missing	✗ Missing	✗ Missing	☑ Included
Ontology-perception duality	✗ Missing	✗ Missing	✗ Missing	☑ Included

7. 结论 / Conclusion

中文

本文系统分析了从纯笛卡尔空间到完整图空间的四个层级：

1. 纯笛卡尔空间 (3D)：最不完整，仅能描述静态位置，缺失时间、频率、相位和全部角动量。它是退化的感知截面。
2. 感知时频相空间 (6D)：人类可读的完整读数界面，包含位置、时间、频率和相位，但作为投影空间丢失了物理本体的非对角耦合分量。
3. 物理角动量空间 (6D)：宇宙物理实在的闭包，由六个实矢量分量 $(L_x, L_y, L_z, S_x, S_y, S_z)$ 张成。
4. 完整图空间 (12D 实空间 = 六维复空间 $\mathbb{C}^6$)：物理本体与感知投影的并置，无损保存全部信息。

$$\mathcal{S}_3^{\text{Cart}} \subset \mathcal{M}_6^{\text{percep}} \subset \mathcal{M}_{12}^{\text{graph}} \supset \mathcal{M}_6^{\text{phys}}$$

核心结论：

笛卡尔空间缺了 9 个实维度。感知空间缺了物理本体的非对角耦合。物理角动量空间由六个实矢量分量张成。完整图空间是六维复空间 $\mathbb{C}^6$ ——三个轨道复数和三个自旋复数——是信息无损的唯一载体。

English

Four hierarchical space structures are derived within the ANG-TOE framework:

1. Pure Cartesian space (3D): the most incomplete level, only describing static position; time, frequency, phase and all angular-momentum information are missing. It is a degenerate perceptual cross-section.
2. Perceptual time-frequency-phase space (6D): the human-readable measurement interface containing position, time, frequency and phase; off-diagonal coupling components of physical ontology are lost in projection.
3. Physical angular-momentum space (6D): the closed ontological manifold of the universe spanned by six real vector components $(L_x, L_y, L_z, S_x, S_y, S_z)$.
4. Full graph space (12-dimensional real space = 6-dimensional complex space $\mathbb{C}^6$): the composite manifold combining physical ontology and perceptual projection, preserving all information without loss.

$$\mathcal{S}_3^{\text{Cart}} \subset \mathcal{M}_6^{\text{percep}} \subset \mathcal{M}_{12}^{\text{graph}} \supset \mathcal{M}_6^{\text{phys}}$$

Core Conclusion:

Cartesian space lacks 9 real dimensions. Perceptual space loses off-diagonal couplings of physical ontology. Physical angular-momentum space is spanned by six real vector components. Full graph space, expressed as the 6-dimensional complex

space $\mathbb{C}^6$ with three orbital complex pairs and three spin complex pairs, is the unique lossless information carrier.

8. 参考文献 / References

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English References

- [1] Song, C. (2026). ANG-TOE v1.6: Ultimate Framework for Universal Computation.
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